DEBAJYOTI MAZUMDER

Ph.D. Scholar · Natural Language Processing

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Summary

Debajyoti Mazumder is a Ph.D. scholar in Data Science and Engineering at IISER Bhopal, working under the guidance of Dr. Akash Anil. His research focuses on code-mixed NLP, a setting within the broader areas of multilingual and low-resource NLP. His Ph.D. focuses on the synergy between code-mixed setting and its constituent native languages. He has published research in transactions such as ACM TALLIP, conferences like EMNLP, WSDM and AACL-IJCNLP, and workshops including DHOW-MiLLA@TheWebConf. His industry experience spans LLM reasoning and reinforcement learning at NielsenIQ and recommender systems at Adobe.

Research Interests

- **Multilingual & Code-Mixed NLP**
- **LLM Interpretability**
- **Representation Learning**
- **Crosslingual and Crossmodal Knowledge Consistency**
- **Automated Fact-Checking**
- **Low-Resource NLP**

Education

Ph.D. in Data Science & Engineering

Aug. 2022 – Present

Indian Institute of Science Education and Research Bhopal (IISER Bhopal) | CPI: 9.00 — *Madhya Pradesh, India*

Supervisor: Dr. Akash Anil. Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Advanced Natural Language Processing, Spatial Data Science & Applications, Data Science in Practice.

M.Sc. in Mathematics

Jul. 2018 – Oct. 2020

Zakir Husain Delhi College, University of Delhi | 69.25% — *New Delhi, India*

B.Sc. (Honours) in Mathematics

Jul. 2015 – Jul. 2018

Deshbandhu College, University of Delhi | SGPA: 8.70 — *New Delhi, India*

Work Experience

Adobe Research

Research Ph.D. Intern — *Jul. 2026 – Sep. 2026*

- Built a next-question recommender system for enterprise (CXO) users.
- Used personalized PageRank to provide context to the recommender system.

NielsenIQ

Research Scientist – Senior Intern — *Jul. 2025 – Jan. 2026*

- Worked on LLM reasoning using reinforcement learning with verifiable rewards (RLVR) for e-commerce tasks.
- Experimented with warm-up supervised fine-tuning and reasoning strategies, including variants of GRPO.

Publications

A Two-Way Street: Surveying the Interplay between Code-Mixing and Native Languages in Text and Speech

D. Mazumder, *A. Anil*

Survey (https://arxiv.org/abs/2610.27631) — *arXiv preprint – 2026*

- Organizes transfer work in both directions, native to code-mixed and the reverse, by direction, route, and modality.
- Finds that native data helps more than new architectures on a shared speech benchmark.

Entailed Opinion Matters: Improving the Fact-Checking Performance of Language Models by Relying on their Entailment Ability

G. Kumar, A. Garg, ***D. Mazumder****, A. Kishore, B. Kumar, J. Patro*

Fact-Checking via Entailment (https://arxiv.org/pdf/2505.15050) — *AACL-IJCNLP Findings – 2026*

- Leverages the entailment ability of language models to improve fact-checking performance.
- Provides a thorough comparison between training-based and prompting-based methods.

To Trust or Not to Trust: Retrieval-Augmented Fact Checking in Speech

D. Mazumder*, Mamta, A. Subramanyam*

Fact-Checking in Speech (https://aclanthology.org/2026.emnlp-main.2534/) — *EMNLP – 2026*

- Studies the transferability of the fact-verification ability of large audio-language models (LALMs) from text to speech.

Evaluating Cross-lingual Knowledge Consistency in Code-Mixed vis-a-vis Indian Languages using IndicKLAR — ***D. Mazumder***, *D. Pathak, P. Kodali, A. Joshi, A. Agarwal, J. Patro*

Cross-Lingual Knowledge Consistency (https://aclanthology.org/2026.findings-emnlp.1655/) — *EMNLP Findings – 2026*

- Introduces IndicKLAR, an 18-language Indic benchmark for cross-lingual knowledge consistency.
- Proposes Translate-in-Thought (TinT) prompting to narrow the gap between low- and high-resource languages.

Neither Here Nor There: Cross-Lingual Representation Dynamics of Code-Mixed Text — A Hindi–English Case Study — ***D. Mazumder***, *D. Pathak, P. Kodali, J. Patro*

Cross-Lingual Representation Dynamics (https://aclanthology.org/2026.findings-emnlp.1748/) — *EMNLP Findings – 2026*

- Introduces a trilingual corpus to probe cross-lingual representation alignment across code-mixed and native languages.
- Proposes a post-training alignment method motivated by interpretability analysis.

On VLMs for Diverse Tasks in Multimodal Meme Classification — *D. Gavit, G. Kumar,* ***D. Mazumder***, *S. Das, J. Patro*

Multimodal Meme Classification (https://dl.acm.org/doi/10.1145/3774905.3796487) — *DHOW-MiLLA @ TheWebConf – 2026*

- A comprehensive study of state-of-the-art vision-language models (VLMs) via prompting and training approaches.
- VLM-understanding-based LLM fine-tuning.

Mind the Links: Cross-Layer Attention for Link Prediction in Multiplex Networks — *D. Sharma, A. Kishore, A. Garg,* ***D. Mazumder***, *D. Mohapatra, J. Patro*

Link Prediction in Multiplex Networks (https://dl.acm.org/doi/10.1145/3773966.3779365) — *ACM WSDM – 2026*

- Introduces link prediction in the partial graph of multiplex networks as a novel task.

Revealing the Impact of Synthetic Native Samples and Multi-Tasking Strategies in Hindi-English Code-Mixed Humour and Sarcasm Detection — ***D. Mazumder***, *A. Kumar, J. Patro*

Multi-Task Humour & Sarcasm Detection (https://aclanthology.org/2025.findings-emnlp.1308/) — *EMNLP Findings – 2025*

- Designs a multi-task learning architecture that leverages shared knowledge across related tasks.
- Outperforms current state-of-the-art models on humour and sarcasm detection.

Improving Code-Mixed Hate Detection by Native Sample Mixing: A Case Study for Hindi-English Code-Mixed Scenario — ***D. Mazumder***, *A. Kumar, J. Patro*

Code-Mixed Hate Detection (https://dl.acm.org/doi/10.1145/3726866) — *ACM TALLIP – 2025*

- Adds native-language hate samples to the training set for code-mixed hate detection.
- Transfers knowledge from native-language hate samples alone to detect code-mixed hate.

Honors & Awards

2026	**Best Paper Award Nominee**, EMNLP 2026, for "To Trust or Not to Trust" (VeriSpeak)	*Budapest, Hungary*
2026	**Student Volunteer and Diversity & Inclusion**, EMNLP 2026	*Budapest, Hungary*
2026	**ACM/IARCS Travel Grant**, EMNLP 2026	*Budapest, Hungary*
2026	**Selected, Machine Learning Summer School (MLSS)**, Columbia University	*New York, USA*
2026	**SIGIR Student Travel Grant**, WSDM 2026	*Boise, Idaho, USA*
2026	**ACM Travel Grant (Invited Talk & Poster Presenter)**, ACM ARCS 2026	*IIT Hyderabad*
2025	**ANRF ITS Travel Grant (Poster Presenter)**, EMNLP 2025	*Suzhou, China*
2025	**Selected, LiveRAG Challenge Track**, SIGIR 2025	*Padua, Italy*
2023	**Travel Grant (Participant)**, Indian Symposium on Machine Learning (IndoML 2023)	*IIT Bombay*
2023	**2nd Position, Poster Presentation**, DSE Day, IISER Bhopal	*Bhopal, India*
2022	**Travel Grant (Participant)**, Indian Symposium on Machine Learning (IndoML 2022)	*IIT Gandhinagar*
2022	**MHRD Fellowship Recipient**, GATE 2022	*India*

Academic Service

Teaching Experience

IISER Bhopal — *Bhopal, India*

Teaching Assistant — *Jan. 2023 – Nov. 2026*

- **Machine Learning with Graphs** (DSE 423/623): Aug.–Nov. 2026
- **Natural Language Processing** (DSE 318/401/607): Jan.–Apr. 2026; (DSE 407/607): Aug.–Nov. 2024, Aug.–Nov. 2023
- **Introduction to Programming** (ECS 102): Jan.–Apr. 2025, Jan.–Apr. 2024
- **Applied Optimization** (DSE 311): Jan.–Apr. 2023

Reviewer Responsibilities

ACL Rolling Review (ARR) Oct. 2026 | ICWSM 2026 | The ACM Web Conference 2026, 2027 | LT-EDI 2026 | DravidianLangTech 2026 | IEEE Transactions on Multimedia | INSTCon 2026